A-A-52017
12 January 1990

COMMERCIAL ITEM DESCRIPTION

COMPRESSOR UNIT: DIESEL-ENGINE-DRIVEN, 450-650 CFM, 350 PSI, WHEEL-MOUNTED

The General Services Administration has authorized the use of this Commercial item description.

This item description covers a commercial, four-wheel-mounted, diesel-engine-driven air compressor. The air compressor shall be complete with all components that are customarily offered with the air compressor of this type, furnished as standard equipment whether or not stipulated herein, together with such other equipment as may be necessary to enable the air compressor to function reliably and efficiently in sustained operation, or to meet requirements set forth in this item description.

Salient characteristics.

Design. The air compressor shall be enclosed by a housing and mounted on a four-wheeled chassis with a tow bar and front axle knuckle and kingpin type steering. The chassis shall be designed for heavy-duty or extreme service. The fuel tank shall have sufficient capacity for 8 hours of normal operation in ambient temperatures ranging from -25 to +100 °F. The fuel tank shall be an integral part of the unit.

Performance. The air compressor shall produce a minimum of 450 cubic feet per minute (CFM) of free air at 350 pounds per square inch, gage (PSIG) in ambient temperatures ranging from -25 to +110 °F.

Engine. The offeror shall furnish the diesel engine with all accessories offered on his commercial air compressors. The diesel engine shall operate on military fuels and lubricants and shall be manufactured in the United States of America. The engine shall conform to all Federal laws and regulations concerning air pollution control in effect for this type of equipment at the time of contract award.

Compressor. The compressor and all accessories shall be as offered on the commercial air compressor and may be of the rotary or reciprocating type.

Regulating device. The regulating device shall be capable of selecting an engine speed and a compressor air intake valve opening to suit any air demand by the air compressor in a smooth, stepless manner.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 4310

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Receiver. The air receiver shall be of sufficient volume to provide a non-pulsating source of air pressure. The receiver shall be provided with a drain valve, a safety relief valve, and an automatic blow-down valve to fully discharge the receiver when the engine stops. The receiver shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code.

Instrumentation. All instruments, gages, and controls supplied commercially as standard equipment shall be supplied on the air compressor.

Manifold. An air service manifold shall be provided with individually controlled 3/4-inch service outlet valves with standard universal pneumatic hose couplings and a 3-inch main outlet controlled by a globe valve.

Brakes. The air compressor shall be provided with a parking brake enabling it to be held in a stationary position while on a 15 percent grade.

Lubrication. The air compressor shall be capable of operating using standard military lubricants. The requirements and procedures of MIL-STD-838 shall be used for lubricant selection and application. All components requiring lubrication shall be provided with a means of lubrication.

Maintenance characteristics. The air compressor design, together with the accessories, shall permit ready access to all items requiring periodic maintenance service in the field. Periodic maintenance shall be accomplished with the use of conventional general purpose tools normally associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the air compressor.

Lifting and tiedown. Lifting and tiedown attachments shall be provided. Instructions for the use of the attachments or procedures shall be included on plates permanently affixed to the air compressor.

Safety. All rotating or reciprocating parts and parts subject to high operational temperatures that are of such a nature or are so located as to be or become a hazard to the operating or attending personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

Noise. The noise level of the air compressor shall meet the requirements of CFR 40, Part 204 - Noise Emission Standards for Construction Equipment.

Treatment and painting. The parts of the air compressor normally painted shall be cleaned, treated, and painted in accordance with the offeror's standard commercial practice. The final color shall be color number X24087 of FED-STD-595.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

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Quality assurance provisions.

Calculations. The offeror shall furnish test data or calculations to show that the air compressor can be operated in ambient temperature ranging from -25 to +110 °F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature and engine oil temperature.

First article demonstration. The first article air compressor produced in accordance with this item description shall be subjected to an operational demonstration consisting of not less than 8 hours of operation at maximum governed engine speed. During the last hour, compressor air delivery at 350 PSIG shall be determined in accordance with ASME Performance Test Code PTC 9. Inability of the air compressor to deliver a minimum of 450 CFM of free air, any malfunction, or any leakage of fuel or lubricants shall be cause for refusal of acceptance of the air compressor by the Government until the problem is corrected. The offeror shall submit a report to the Government not less than 15 days upon conclusion of the demonstration. This report shall include a description and results of the test performed, including any malfunctions and actions taken to correct such malfunctions.

Run-in. Each air compressor (except first article) shall be subjected to the manufacturer's standard run-in procedure, except that the run-in shall be for not less than two hours of operation. Any malfunction or any leakage of fuels or lubricants shall be cause for refusal of acceptance of the air compressor by the Government until the problem is corrected.

Packaging and marking. Military packaging shall be specified when shipments are to overseas Department of Defense (DoDO consignees. Commercial packaging may be specified for shipments to DoD consignees within the continental United States and to non-DoD consignees worldwide. When a supplier's normal packaging will satisfy the requirements for military packaging, it may be used.

Military packaging. Each compressor and its accessories shall be preserved and packed to protect against physical damage and deterioration in logistic environment characterized by the following:

1. Multiple handling during transportation and in-transit storage.
2. Shock, vibration and static loading forces during worldwide shipment by all transport modes.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and in-transit transfers.
5. Stacking loads during shipment and storage.

The engine shall be internally preserved. Quantities of compressors destined to a single address may be utilized into one or more unit loads.

Commercial. Each compressor shall be packed in accordance with the supplier's normal method for retail distribution. Compressors shipped to a single destination may be unitized to permit handling as a single unit. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification rules.

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Marking. Unit, intermediate (when provided), and exterior packs shall be marked with the item nomenclature, stock or part number, quantity per pack, and contract number as a minimum: additional marking shall be as specified in the contract or order. Exterior packs shall have, in addition, the complete consignee's address. Marking shall be applied in a legible manner, in a contrasting color, of an appropriate size, and of waterproof materials.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Notes. The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. When military packing is required.
3. When additional marking is required.
4. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.

FED-STD-595 and MIL-STD-838 are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

The ASME Boiler and Pressure Vessel Code and the ASME Performance Test Code PTC 9 are available from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

CFR 40, Part 204 is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402

National Motor Freight Classification is available from the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.

Uniform Freight Classification is available from the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.

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